

BM, BMhp

Installation and operating instructions



CONTENTS

	Page
1. Symbols used in this document	2
2. General description	2
2.1 Pumped liquids	2
2.2 Sound pressure level	2
3. Delivery, transportation and storage	2
3.1 Frost protection	2
4. Preparation	3
5. Installation	3
5.1 Booster modules connected in series and in parallel	4
6. Pipe connection	4
7. Electrical connection	5
7.1 Frequency converter operation	5
8. Motor protection	6
8.1 Setting of motor-protective circuit breaker	6
9. Before starting the booster module(s)	6
10. Operation	6
10.1 Limitations to operation	7
11. Automatic monitoring devices	7
12. Checking of operation	7
13. Fault finding	8
14. Service documentation	8
15. Checking of motor and cable	9
16. Disposal	9

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

**Warning**

If these safety instructions are not observed, it may result in personal injury.

**Warning**

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General description

Grundfos BM and BMhp booster modules are designed for pressure boosting, liquid transfer and circulation in systems with a high static pressure.

2.1 Pumped liquids

Thin, non-explosive liquids, not containing abrasive particles or fibres. The liquid must not attack the pump materials chemically and mechanically.

**Warning**

The booster modules must not be used for the pumping of flammable liquids such as diesel oil, petrol or similar liquids.

2.2 Sound pressure level

The sound pressure level of BM 4" and BM and BMhp 6" booster modules is lower than 70 dB(A).

The sound pressure level of BM and BMhp 8" booster modules is lower than 80 dB(A).

3. Delivery, transportation and storage

Delivery

The booster modules are supplied from the factory in proper packing in which they should remain until they are to be installed. The modules are ready for installation.

Transportation and storage

During transportation and storage, the booster modules may only be placed in the positions shown in fig. 1.

Before storage, the booster modules should be flushed through with clean freshwater. See section [10.1.2 Flushing of the booster module](#).

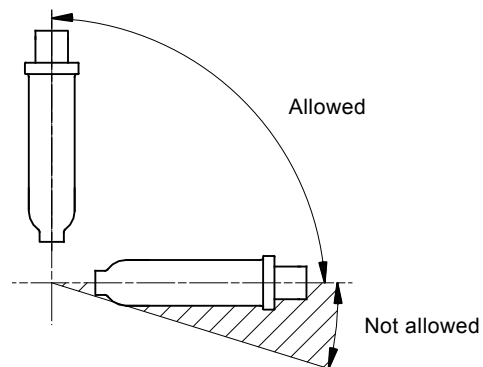


Fig. 1 Positional requirements

Caution If the module has been stored for more than one year, the motor liquid must be checked and refilled, if required.

Motors of standard modules are factory-filled with a Grundfos motor liquid, SML-3, which is frost-proof down to -20 °C.

Motors of modules in special version may be filled with demineralised water, i.e. not frost-proof.

3.1 Frost protection

If the module has to be stored, it must be stored on a frost-free location, or it must be ensured that the motor liquid is frost-proof.

4. Preparation

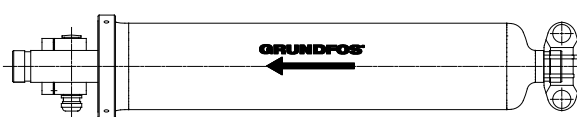
Before installation, the following checks should be made:

- Check for transport damages**
Make sure that the module has not been damaged during transportation.
- Type of module**
Check that the type designation given on the nameplate fitted to the sleeve corresponds to order.
- Power supply**
The motor voltage and frequency are marked on the nameplate. Make sure that the motor is suitable for the power supply on which it will be used.
- Liquid in motor**
If a module has been stored for more than one year, check the motor liquid and refill, if required. Contact Grundfos. If a module is supplied for a special system, the motor may be supplied without liquid or with demineralised water. See section 3. *Delivery, transportation and storage*.

Versions

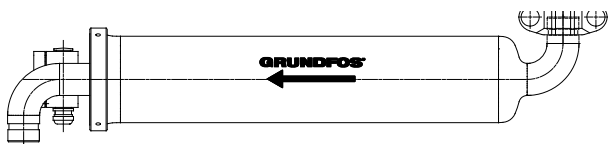
BM 4"

Straight version



TM00 3793 4698

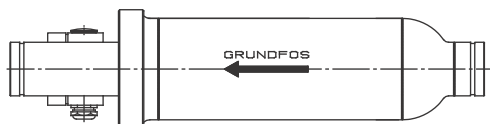
Bent version



TM00 3794 4698

BM/BMhp 6"

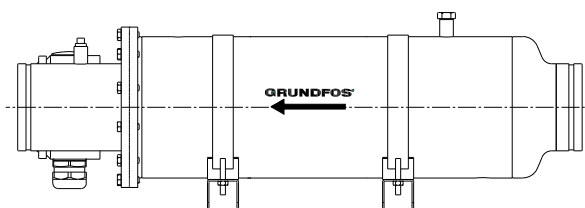
BM/BMhp 6" is only available in straight version.



TM00 4019 2410

BM/BMhp 8"

BM/BMhp 8" is only available in straight version.



TM01 1420 1912

Fig. 2 Flow directions

5. Installation

Grundfos booster modules are as standard supplied without a built-in non-return valve. However, a non-return valve can be built in on request. See fig. 7.

In systems involving the risk of water hammer in connection with start/stop, the necessary measures must be taken to minimise this risk.

The booster modules are suitable for both vertical and horizontal installation, however, the discharge port should never fall below the horizontal plane. See fig. 3.

An arrow on the module sleeve shows the direction of flow of liquid through the module. See fig. 2.

The module is fastened by means of clamps. See fig. 5.



Warning

Note that the module has an uneven weight distribution. Because of the motor, the largest weight will be in the first third of the sleeve (when seen from the suction port).



Warning

Observe local regulations concerning limits for manual lifting or handling.

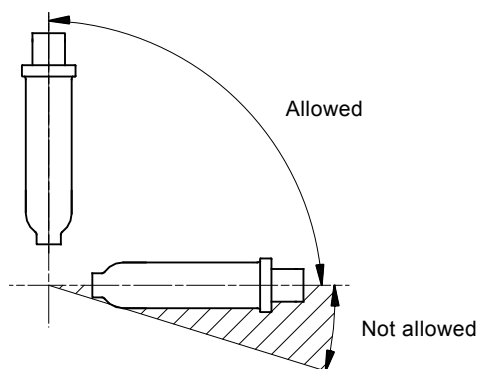


Fig. 3 Positional requirements

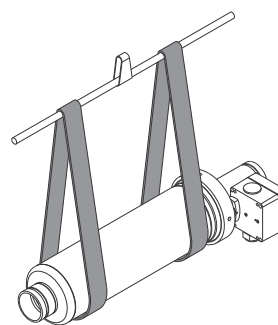


Fig. 4 How to lift the pump

Caution Do not lift the pump by terminal box.

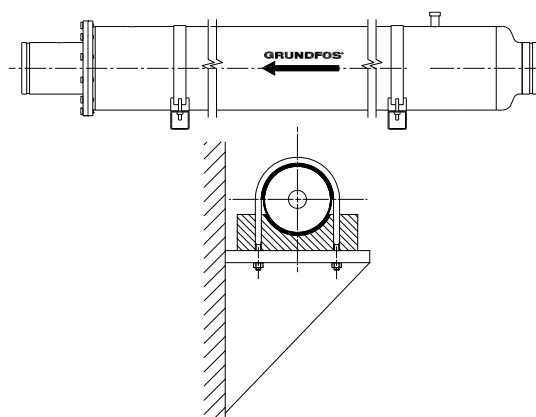


Fig. 5 Fastening by means of clamps

TM01 1282 4197

TM06 6892 2616

TM02 5911 4002 - TM00 4041 1912

5.1 Booster modules connected in series and in parallel

- For modules connected in series, mounted above each other, we recommend to connect the pipes as shown in fig. 6.
- For modules connected in parallel, mounted above each other, we recommend to connect the pipes as shown in fig. 7. This layout ensures that the modules are filled with water before starting.
- For modules connected in series and parallel, mounted above each other, we recommend to connect the pipes as shown in fig. 8.
- For modules connected in series and fitted with a bypass, mounted above each other, we recommend to connect the pipes as shown in fig. 9.

Symbols used in figures 6 to 9:

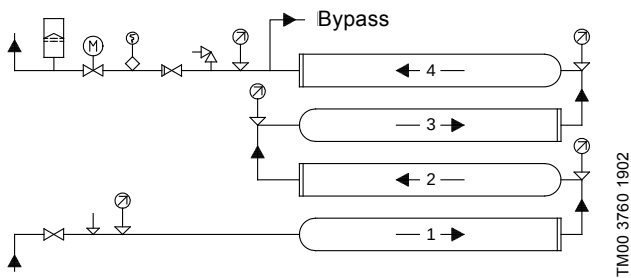
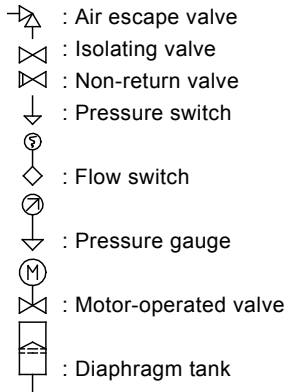


Fig. 6 Booster system with four modules connected in series, mounted above each other

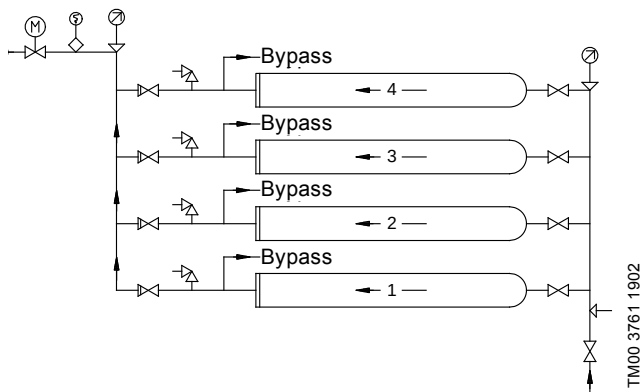


Fig. 7 Booster system with four modules connected in parallel, mounted above each other

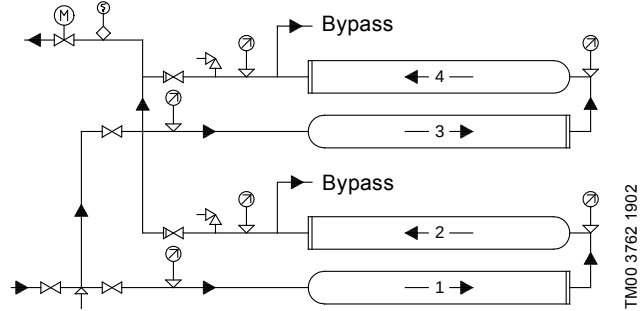


Fig. 8 Booster system with two modules connected in series and in parallel, mounted above each other

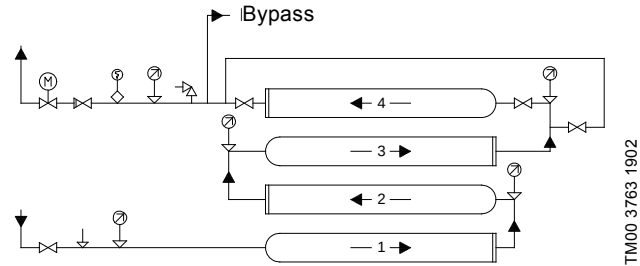


Fig. 9 Booster system with four modules connected in series with bypass, mounted above each other

Note As venting problems may arise in such installations, it is advisable to install suitable vent devices.

6. Pipe connection

The booster modules are fitted with clamp liners for Victaulic/PJE clamp couplings on the suction and discharge sides. See fig. 10.

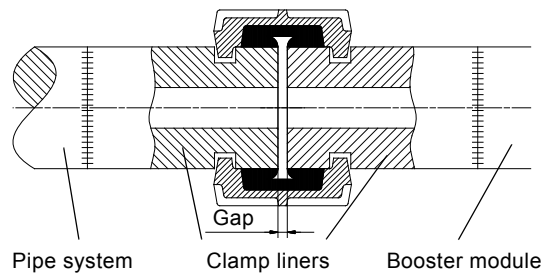


Fig. 10 Pipe connection using clamp couplings

Caution Avoid stress in the pipe system.

Note Clean Victaulic connections, rubber and pipes, with freshwater in order to prevent crevice corrosion.

Size	Type	Victaulic coupling	Gap [mm]
BM 4"	BM 3A - BM 9	1 1/4" Ø42	1
BM/BMhp 6"	BM 17 - BM 60	3" Ø89	3
BM 8"	BM 30 - BM 46	3" Ø89	3
BM 8"	BM 60	4" Ø114	6
BM/BMhp 8"	BM 77 - BM 95	5" Ø139	6
BM/BMhp 8"	BM 125 - BM 215	6" Ø168	6

7. Electrical connection

The electrical connection must be carried out by an authorised electrician in accordance with local regulations.

Warning



Connect the pump to an external mains switch placed close to the pump and to a motor-protective circuit breaker.

It must be possible to lock the mains switch in OFF position, isolated. Type and requirements as specified in EN 60204-1, 5.3.2.

Warning



Before making any electrical connections, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

The pump must be earthed.

The required voltage quality measured at the module terminals is - 10 %/+ 6 % of the rated voltage during continuous operation (including variation in the supply voltage and losses in cables). Check that there is voltage symmetry in the power supply lines, i.e. approximately same difference of voltage between the individual phases. See also section 15. *Checking of motor and cable*, point 1.

In order that the modules meet the EC EMC Directive (2004/108/EC), a 0.47 µF capacitor (in accordance with IEC 384-14) must always be connected over the two phases to which the temperature transmitter is connected. See fig. 11.

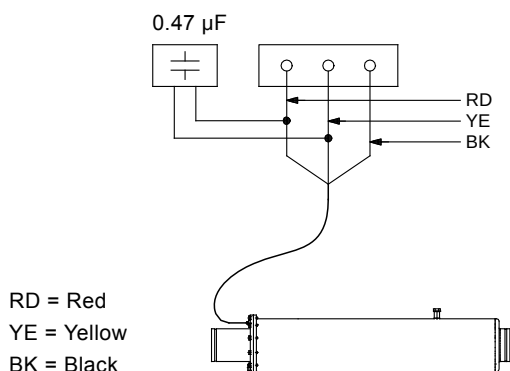


Fig. 11 Electrical connection

The electrical connections must be made close to the flange (fig. 12), either by means of a terminal box (figures 13 and 14) or a cable termination kit.

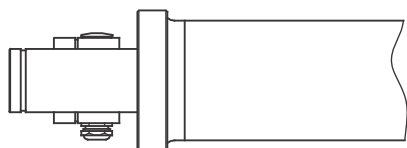


Fig. 12 BM/BMhp with terminal box

BM 4", BM/BMhp 6", BM/BMhp 8"

Direct-on-line

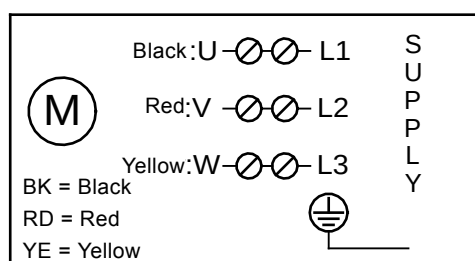


Fig. 13 Wiring diagram, DOL connection

BM/BMhp 6", BM/BMhp 8"

Star-delta starting

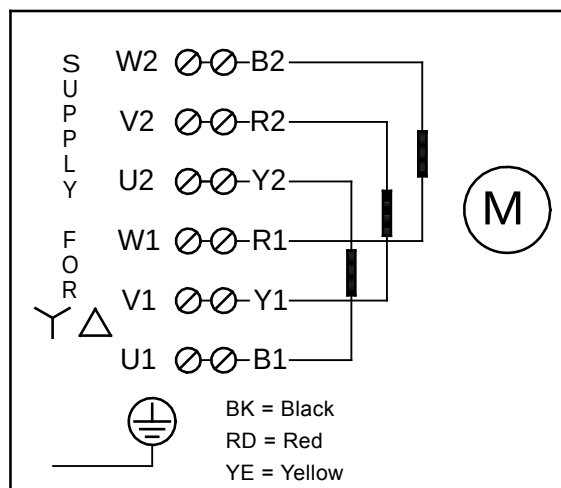


Fig. 14 Wiring diagram, star-delta connection

7.1 Frequency converter operation

Grundfos motors

Three-phase Grundfos MS and MMS motors can be connected to a frequency converter.

If a frequency converter is to be used for booster modules connected in series, it must be connected to the last module in the flow direction.

We recommend Grundfos CUE. The CUE is a series of frequency converters designed for speed control of a wide range of Grundfos products, for example BM and BMhp.

If a Grundfos MS motor with a temperature transmitter is connected to a frequency converter, a fuse incorporated in the transmitter will melt and the transmitter will be inactive. This means that the motor will operate like a motor without a temperature transmitter.

Caution

During frequency converter operation, it is not advisable to run the motor at a frequency higher than the rated frequency (50 or 60 Hz). In connection with pump operation, it is important never to reduce the frequency, and consequently the speed, to such a level that the necessary flow of cooling liquid past the motor is no longer ensured.

The permissible frequency ranges are 30-50 Hz and 30-60 Hz, respectively.

During start, the maximum acceleration time from 0 to 30 Hz is 3 seconds.

During stop, the maximum deceleration time from 30 to 0 Hz is 3 seconds.

Depending on the frequency converter type, it may expose the motor to detrimental voltage peaks.

The above disturbance can be abated by installing an RC filter between the frequency converter and the motor.

Possible increased acoustic noise from the motor can be abated by installing an LC filter which will also eliminate voltage peaks from the frequency converter.

For further details, please contact your frequency converter supplier or Grundfos.

TM00 4034 3197

TM02 5255 2402

TM02 5256 2410

TM00 4035 1694

8. Motor protection

The booster module must be connected to an effective motor-protective circuit breaker which must protect the motor against damage from voltage drop, phase failure, overload and a locked rotor.

In power supply systems where undervoltage and variations in phase symmetry may occur, a phase failure relay should also be fitted. See section [15. Checking of motor and cable](#).

The control circuit must always be made in such a way that all modules are stopped if one module fails.

8.1 Setting of motor-protective circuit breaker

For cold motors, the tripping time for the motor-protective circuit breaker must be less than 10 seconds (Class 10) at 5 times the rated full-load current ($I_{1/1}/I_{SFA}$) of the module. See the module nameplate.

Note

If this requirement is not met, the motor warranty will be invalidated.

In order to ensure the optimum protection of the submersible motor, the motor-protective circuit breaker should be set in accordance with the following guidelines:

1. Set the motor-protective circuit breaker to the rated full-load current ($I_{1/1}/I_{SFA}$) of the module.
2. Start the module, and let it run for half an hour at normal performance. See section [9. Before starting the booster module\(s\)](#).
3. Slowly grade down the scale indicator until the motor trip point is reached.
4. Increase the overload setting by 5 %.

The highest permissible setting is the rated full-load current ($I_{1/1}/I_{SFA}$) of the module.

For modules wound for star-delta starting, the motor-protective circuit breaker should be set as described above, but the maximum setting should be as follows:

Motor-protective circuit breaker setting = rated full-load current x 0.58.

The highest permissible startup time for star-delta starting is 2 seconds.

9. Before starting the booster module(s)

The booster module(s) must be filled with water before startup.

Procedure:

1. Slacken the vent valve of the booster module, if any.
2. Ensure an inlet pressure on the booster module.
3. Completely open the isolating valve.
4. Wait 3 to 5 minutes to ensure optimum venting.
5. Tighten and clean the vent valve.

Checking the direction of rotation

Procedure:

1. Close the isolating valve on the discharge side of module 1 (figures [6](#) to [9](#)) to approx. 1/3 of maximum flow.
2. Start module 1, and record discharge pressure and flow readings.
3. Stop the module, and interchange two of the phases to the module.
4. Restart the module, and re-record discharge pressure and flow readings.
5. Stop the module.
6. Compare the results taken under points 2 and 4. The connection which gave the larger pressure and flow is the correct direction of rotation.

The check of the direction of rotation should last for the shortest possible time.

If the booster system has several modules, starting and rotation checks must be made in the order 1-2-3-4 until all modules are running. When module 2 is checked, module 1 must be running. When module 3 is checked, modules 1 and 2 must be running, etc.

If modules in series are also connected in parallel (fig. [8](#)), the direction of rotation of each section connected in series should be checked.

After having checked the direction of rotation, stop the modules in the order 4-3-2-1.

The booster system is now ready for operation.

10. Operation

Start

Warning

The booster modules are not allowed to run against a closed discharge valve for more than 5 seconds as this will cause an increase in temperature/formation of steam in the module which may cause damage to the pump and motor.



If there is any risk of the modules running against a closed discharge valve, a minimum liquid flow through the module should be ensured by connecting a bypass or a drain to the discharge side of the module. The drain can for instance be connected to a tank.

The booster modules must be started in the order 1-2-3-4 at intervals of 1 to 2 seconds. See section [5.1 Booster modules connected in series and in parallel](#). If other intervals are required, contact Grundfos.

Module 1 is always the first module on the suction side. During startup, we recommend to close the isolating valve 3/4 and open it slowly (2 to 3 seconds).

In systems involving the risk of water hammer in connection with start/stop, the necessary measures must be taken to minimise this risk, e.g. by installing a diaphragm tank.

Operation

During operation, the inlet pressure must be checked in accordance with section [10.1 Limitations to operation](#).

The total discharge pressure and flow of modules connected in series should never be changed by stopping one or more of the modules. If other pressures or flows are required, the following procedures are applicable:

- Bypass of module(s): Install a bypass between two modules. See fig. [9](#).
Stop the module(s) which is/are not required, and close the isolating valves on either side of the module. See section [10.1.2 Flushing of the booster module](#).
The module(s) to be bypassed is/are always the last module(s) in the flow direction.
- Fit a reducing valve to the discharge pipe.
The values stated in section [10.1 Limitations to operation](#) must be observed.
- Modules with three-phase motors: Install a frequency converter for speed control of the last booster module in the flow direction. See section [7.1 Frequency converter operation](#).

Stop

The modules must be stopped in reverse order, i.e. 4-3-2-1, at intervals of 1 to 2 seconds. See section [5.1 Booster modules connected in series and in parallel](#).

If the booster system is taken out of operation for a long period, the modules should be flushed through with clean freshwater. See section [10.1.2 Flushing of the booster module](#). The modules are then left with freshwater until they are to be used again.

10.1 Limitations to operation

The flow limits stated in the table below apply to the optimum operating ranges of the modules as regards efficiency.

Recommended flow rate at 25 °C / 77 °F				
Type	m ³ /h		US gpm	
	50 Hz	60 Hz	50 Hz	60 Hz
BM 3A	0.8 - 4.4	1.0 - 4.7	3.5 - 20	4.4 - 21
BM 5A	2.5 - 6.8	3.0 - 7.7	11-30	13-34
BM 9	4.0 - 11	4.8 - 11	17-48	21-48
BM/BMhp 17	8-22	10-26	35-176	44-115
BM/BMhp 30	15-38	19-45	66-167	84-198
BM/BMhp 46	24-60	28-72	106-264	123-317
BM/BMhp 60	30-75	37-90	132-330	163-396
BM/BMhp 77	38-96	47-120	167-422	207-528
BM/BMhp 95	47-118	57-143	207-520	251-629
BM/BMhp 125	62-156	75-187	273-686	330-823
BM/BMhp 160	78-195	90-215	343-858	396-946
BM/BMhp 215	98-265	115-310	431-1166	506-1364

The pressure limits stated in the table below must be observed.

Recommended pressure ¹⁾						
Type	Inlet pressure				Outlet pressure	
	Min.		Max. ²⁾		Max. ²⁾	
	[bar]	[psi]	[bar]	[psi]	[bar]	[psi]
BM 4"	0.5	7.25	60	870	80	1160
BM 6"	0.5	7.25	50	725	80	1160
BM 8"	1	14.5	25	362	70	1015
BMhp 6"	0.5	7.25	80	725	80	1160
BMhp 8"	1	14.5	80	362	80	1015

1) Contact Grundfos if higher pressures are required.

2) **Note:** If the maximum inlet/outlet pressure is exceeded, install a safety valve.

10.1.1 Cooling

The temperature and flow limits stated in the table below must be observed to ensure sufficient cooling of the motor.

Maximum permissible liquid temperature				
Motor	Maximum liquid temperature		Minimum flow velocity past the motor	Minimum flow
	[°C]	[°F]	[m/s (ft/s)]	[m ³ /h (US GPM)]
Grundfos 4"	40	104	≥ 0.15 (0.49)	≥ 0.8 (3.5)
Grundfos 6"	40	104	≥ 0.15 (0.49)	≥ 5.5 (24)
Grundfos 8"	40	104	≥ 0.15 (0.49)	≥ 18.5 (81.5)

10.1.2 Flushing of the booster module

When a module is flushed, the flow must be reduced to maximum 10 % of the rated flow at a minimum pressure of 2 bar. The modules must be stopped while the system is flushed. See section [10. Operation](#).

Note

When pumping water with a salinity above 2000 ppm NaCl, the module must be flushed through in the flow direction as described in the following.

The flushing procedure depends on the operating condition:

- Intermittent operation**

If the booster module has to be stopped for more than 30 minutes, it must be flushed through with clean freshwater for approx. 10 minutes. The flushing must be continued until the module is completely filled with clean freshwater. The salinity of the flush water must be below 500 ppm on the discharge side.

- Continuous**

- operation**

BM: Once every 24 hours, the module should be stopped and flushed through with the pumped liquid by means of the feed pump.

BMhp: Due to an internal bypass, it is not necessary to stop the BMhp booster modules every 24 hours.

- Interruption of operation**

In case of power, pump or motor failure, the booster module must be cooled and flushed through with clean freshwater. The drain plug can be used when flushing the booster module. The booster module can also be disconnected, removed, dismantled and cleaned with clean freshwater.

10.1.3 Frequency of starts and stops

Minimum 1 per year is recommended.

Maximum 20 per hour.

Maximum 100 per day.

Caution

8" modules: Maximum 20 per day.

11. Automatic monitoring devices

To protect the modules against dry running and to ensure a minimum flow of cooling water past the motors, the system must be fitted with flow and pressure monitoring devices (figures 6 to 9).

The pressure switch on the suction side is set in accordance with the estimated inlet pressure. At a pressure lower than 0.5 bar for BM 4" and BM/BMhp 6", and 1 bar for BM/BMhp 8", an alarm is given and the module must be stopped without delay.

All discharge connections to the system should be fitted with a flow switch which will stop the system at the set minimum flows.

The above monitoring devices ensure a correct inlet pressure and a minimum flow of cooling water past the motor.

If the modules are stopped automatically, automatic flushing is recommended. See section [10.1.2 Flushing of the booster module](#).

12. Checking of operation

Depending on the number of operating hours of the modules, the following should be checked at suitable intervals:

- flow
- starting frequency
- control and protective devices
- liquid temperature
- minimum flow through modules during operation.

If any of the above checks reveal any abnormal operating details, inspection should be carried out in accordance with section [13. Fault finding](#).

We recommend to use the log book at the end of these instructions.

13. Fault finding



Warning

Before removing the terminal box cover and before removing/dismantling the module, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

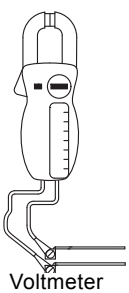
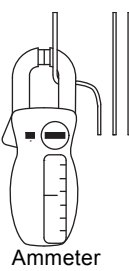
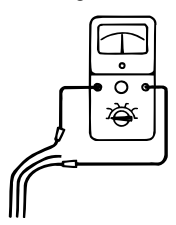
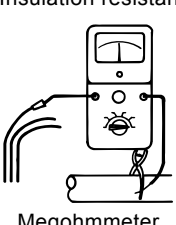
Fault	Cause	Remedy
1. The booster system stops occasionally. One or more modules stop.	a) No or insufficient water supply. The pressure switch has cut out.	Check that the pressure switch functions normally (without delay) and is adjusted correctly. Check that the minimum inlet pressure is correct.
	b) The capacity is too small. The flow switch has cut out.	The discharge pipe is totally or partly blocked due to incorrectly adjusted manually operated valve or failure in the magnetic or motor-operated valve. Check these valves. The flow switch is faulty or incorrectly adjusted. Check/adjust the switch.
2. The booster system does not run.	a) The fuses are blown.	Check and replace both main fuses and/or fuses for control circuit.
	b) The motor-protective circuit breaker has tripped.	Reset the motor-protective circuit breaker. If it trips again, the voltage should be checked.
	c) The magnetic coil of motor-protective circuit breaker/contactors is short-circuited (not cutting in).	Replace the coil. Check the coil voltage.
	d) The control circuit has cut out or is defective.	Check the control circuit and the contacts in the monitoring devices (pressure switch, flow switch, etc.).
	e) The motor/supply cable is defective.	Check motor and cable. See also section 15. Checking of motor and cable .
3. The booster system runs, but gives no water or develops no pressure.	a) No water or too small quantity delivered to the modules or air in the system.	Check that the inlet pressure during operation is at least 0.5 bar for BM 4" and BM/BMhp 6", and 1 bar for BM/BMhp 8". If this is the case, the water supply is OK. Stop and vent the system. Restart the system as described in section 10. Operation . If the module is defective, it should be dismantled and repaired or replaced.
	b) Suction parts are blocked.	Pull the pump out of the sleeve, and clean the suction parts.
4. The booster system runs at reduced capacity (flow and pressure).	a) Wrong direction of rotation.	See section 9. Before starting the booster module(s) .
	b) The valves on the discharge side are partly closed or blocked.	Check valves.
	c) The discharge pipe is partly blocked by impurities.	Measure the discharge pressure and compare with the calculated data. Clean or replace the discharge pipe.
	d) The module is partly blocked by impurities.	Pull the pump out of the sleeve. Dismantle, clean and check the pump. Replace defective parts. Clean the pipes.
	e) The module is defective.	Pull the pump out of the sleeve. Dismantle, clean and check the pump. Replace defective parts. See section 14. Service documentation .

14. Service documentation

Service documentation is available in Grundfos Product Center:
<http://product-selection.grundfos.com/>.

If you have any questions, please contact the nearest Grundfos company or service workshop.

15. Checking of motor and cable

1. Supply voltage  Voltmeter TM00 1371 5092	Measure the voltage between the phases by means of a voltmeter. Connect the voltmeter to the terminals in the motor-protective circuit breaker.	The voltage should, when the motor is loaded, be within - 10 %/+ 6 % of the rated voltage. In case of larger variations in voltage, the motor may burn. If the voltage is constantly too high or too low, the motor must be replaced by one corresponding to the supply voltage. Large variations in voltage indicate poor power supply, and the module should be stopped until the fault has been found. Resetting of the motor-protective circuit breaker may be necessary.
2. Current consumption  Ammeter TM00 1372 5092	Measure the current of each phase while the module is operating at a constant discharge pressure (if possible at the capacity where the motor is most heavily loaded). For normal operating current, see nameplate.	Calculate the current unbalance (%) in the three phases as follows: 1. Add up the three phase currents. 2. Divide this value by three to ascertain the average current. 3. Find the phase current differing most from the average current. 4. Compare the results from points 2 and 3. 5. Divide the difference by the average value and multiply by 100. The result is the current unbalance in %. For three-phase motors, the current unbalance must not exceed 5 %. If so, or if the current exceeds the maximum operating current, there are the following possible faults: • The contacts of the motor-protective circuit breaker are burnt. Replace the contacts. • Poor connection in leads, possibly in terminal box. • Too high or too low supply voltage. See point 1. • The motor windings are short-circuited or partly disjointed. See point 3. • Damaged pump is causing the motor to be overloaded. Pull the pump out of the sleeve for overhaul. • The resistance value of the motor windings deviates too much. Move the phases in phase order to a more uniform load. If this does not help, see point 3.
Items 3 and 4: Measurement not necessary if supply voltage and current consumption are normal.		
3. Winding resistance  Ohmmeter TM00 1373 5092	Remove the phase leads from the terminal box. Measure the winding resistance as shown on the drawing.	The highest value must not exceed the lowest value by more than 10 %. If the deviation is higher, pull the pump out of the sleeve. Measure motor and cable separately, and replace or repair defective parts. See section 14. Service documentation.
4. Insulation resistance  Megohmmeter TM00 1374 5092	Remove the phase leads from the terminal box. Measure the insulation resistance from each phase to earth (frame). Make sure that the earth connection is made carefully.	If the measured insulation resistance is less than 0.5 MΩ and the supply cable is OK, pull the pump out of the sleeve for motor or cable repair or replacement. See section 14. Service documentation.

16. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

System sketch

Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products BM and BMhp, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

CZ: ES prohlášení o shodě

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky BM a BMhp, na něž se toto prohlášení vztahuje, jsou v souladu s ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství v oblastech:

DK: EF-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne BM og BMhp som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

ES: Declaración CE de conformidad

Nosotros, Grundfos, declaramos bajo nuestra entera responsabilidad que los productos BM y BMhp, a los cuales se refiere esta declaración, están conformes con las Directivas del Consejo en la aproximación de las leyes de los Estados Miembros del EM:

FR: Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits BM et BMhp, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

HR: EZ izjava o usklađenosti

Mi, Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod BM i BMhp, na koji se ova izjava odnosi, u skladu s direktivama ovog Vijeća o usklađivanju zakona država članica EU:

IT: Dichiarazione di conformità CE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti BM e BMhp, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

LV: EK atbilstības deklarācija

Sabiedrība GRUNDFOS ar pilnu atbildību dara zināmu, ka produkti BM un BMhp, uz kuriem attiecas šīs paziņojums, atbilst šādām Padomes direktīvām par tuvināšanos EK dalībvalstu likumdošanas normām:

PL: Deklaracja zgodności WE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze wyroby BM oraz BMhp, których deklaracja niniejsza dotyczy, są zgodne z następującymi wytycznymi Rady d/s ujednolicenia przepisów prawnych krajów członkowskich WE:

RO: Declarație de conformitate CE

Noi, Grundfos, declarăm pe propria răspundere că produsele BM și BMhp, la care se referă această declarație, sunt în conformitate cu aceste Directive de Consiliu asupra armonizării legilor Statelor Membre CE:

SE: EG-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna BM och BMhp, som omfattas av denna försäkran, är i överensstämmelse med rådets direktiv om inbördes närmande till EU-medlemsstaternas lagstiftning, avseende:

SK: Prehlásenie o konformite ES

My firma Grundfos prehlasujeme na svoju plnú zodpovednosť, že výrobky BM a BMhp, na ktoré sa toto prehlásenie vzťahuje, sú v súlade s ustanovením smernice Rady pre zblíženie právnych predpisov členských štátov Európskeho spoločenstva v oblastiach:

BG: EC декларация за съответствие

Ние, фирма Grundfos, заявяваме с пълна отговорност, че продуктите BM и BMhp, за които се отнася настоящата декларация, отговарят на следните указания на Съвета за уеднавяване на правните разпоредби на държавите членки на ЕС:

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte BM und BMhp, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen:

EE: EL vastavusdeklaratsioon

Meie, Grundfos, deklareerime enda ainuvastutusel, et tooted BM ja BMhp, mille kohta käesolev juhend käib, on vastavuses EÜ Nõukogu direktiividega EMÜ liikmesriikide seaduste ühitamise kohta, mis käsitlevad:

FI: EY-vaatimustenmukaisuusvakuutus

Me, Grundfos, vakuutamme omalla vastuullamme, että tuotteet BM ja BMhp, joita tämä vakuutus koskee, ovat EY:n jäsenvaltioiden lainsäädännön yhdenmukaistamiseen tähtävien Euroopan neuvoston direktiivien vaatimusten mukaisia seuraavasti:

GR: Δήλωση συμμόρφωσης EC

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα BM και BMhp στα οποία αναφέρεται η παρούσα δήλωση, συμμορφώνονται με τις εξής Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΕ:

HU: EK megfeleléségi nyilatkozat

Mi, a Grundfos, egyedüli felelősséggel kijelentjük, hogy a BM és BMhp termékek, amelyekre jelen nyilatkozik vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak:

LT: EB atitikties deklaracija

Mes, Grundfos, su visa atsakomybe pareiškiamė, kad gaminiai BM ir BMhp, kuriems skirta ši deklaracija, atitinka šias Tarybos Direktyvas dėl Europos Ekonominės Bendrijos šalių narių įstatymų suderinimo:

NL: EC overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten BM en BMhp waarop deze verklaring betrekking heeft, in overeenstemming zijn met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG Lidstaten betreffende:

PT: Declaração de conformidade CE

A Grundfos declara sob sua única responsabilidade que os produtos BM e BMhp, aos quais diz respeito esta declaração, estão em conformidade com as seguintes Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE:

RS: EC deklaracija o usaglašenosti

Mi, Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod BM i BMhp, na koji se ova izjava odnosi, u skladu sa direktivama Saveta za usklađivanje zakona država članica EU:

SI: ES izjava o skladnosti

V Grundfosu s polno odgovornostjo izjavljamo, da so naši izdelki BM in BMhp, na katere se ta izjava nanaša, v skladu z naslednjimi direktivami Sveta o približevanju zakonodaje za izenačevanje pravnih predpisov držav članic ES:

UA: Декларация відповідності ЄС

Компанія Grundfos заявляє про свою виключну відповідальність за те, що продукти BM та BMhp, на які поширюється дана декларація, відповідають таким рекомендаціям Ради з уніфікації правових норм країн - членів ЄС:

AR: إقرار المطابقة من EC (المجموعة الأوروبية)

نقرر نحن، جرونډفوس، بمقتضى مسؤوليتنا الفردية بأن المنتجات BM وBMhp، التي يختص بها هذا الإقرار، تكون مطابقة لتوجيهات المجلس بشأن التقريب بين قوانين الدول أعضاء المجموعة الأوروبية (EC):

- Machinery Directive (2006/42/EC).
Standard used: EN 809:1998 + A1:2009.
- EMC Directive (2014/30/EU)
Standard used: EN 60034-1:2010.
Applies only to pumps with Tempcon sensors.
- Ecodesign Directive (2009/125/EC).
Water pumps:
Commission Regulation No 547/2012.
Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 150095 1116)

Bjerringbro, 9th November 2016



Carsten Høybye Pedersen
Senior Manager
Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark

Person authorised to compile the technical file and
empowered to sign the EC declaration of conformity.

RUS

BM, BMh**Руководство по эксплуатации**

Руководство по эксплуатации на данное изделие является составным и включает в себя несколько частей:

Часть 1: настоящее «Руководство по эксплуатации».

Часть 2: электронная часть «Паспорт. Руководство по монтажу и эксплуатации» размещенная на сайте компании Грундфос. Перейдите по ссылке, указанной в конце документа.

Часть 3: информация о сроке изготовления, размещенная на фирменной табличке изделия.

Сведения о сертификации:

Насосы типа BM, BMh сертифицированы на соответствие требованиям Технических регламентов Таможенного союза: ТР ТС 004/2011 «О безопасности низковольтного оборудования»; ТР ТС 010/2011 «О безопасности машин и оборудования»; ТР ТС 020/2011 «Электромагнитная совместимость технических средств».

KAZ

BM, BMh**Пайдалану бойынша нұсқаулық**

Атаулы өнімге арналған пайдалану бойынша нұсқаулық құрамалы болып келеді және келесі бөлімдерден тұрады:

1 бөлім: атаулы «Пайдалану бойынша нұсқаулық»

2 бөлім: Грундфос компаниясының сайтында орналасқан электронды бөлім «Төлқұжат, Құрастыру және пайдалану бойынша нұсқаулық». Құжат соңында көрсетілген сілтеме арқылы өтіңіз.

3 бөлім: өнімнің фирмалық тақташасында орналасқан шығарылған уақыты жөніндегі мәлімет

Сертификаттау туралы ақпарат:

BM, BMh типті сорғылары «Төмен вольтты жабдықтардың қауіпсіздігі туралы» (ТР ТС 004/2011), «Машиналар және жабдықтар қауіпсіздігі туралы» (ТР ТС 010/2011) «Техникалық заттардың электрлі магниттік сәйкестілігі» (ТР ТС 020/2011) Кеден Одағының техникалық регламенттерінің талаптарына сәйкес сертификатталды.

KG

BM, BMh

Пайдалануу боюнча колдонмо

Аталган жабдууну пайдалануу боюнча колдонмо курамдык жана өзүнө бир нече бөлүкчөнү камтыйт:

1-Бөлүк: «Пайдалануу боюнча колдонмо»

2-Бөлүк: «Паспорт. Пайдалануу жана монтаж боюнча колдонмо» электрондук бөлүгү Грундфос компаниянын сайтында жайгашкан. Документтин аягында көрсөтүлгөн шилтемеге кайрылыңыз.

3-Бөлүк: жабдуунун фирмалык тактасында жайгашкан даярдоо мөөнөтү тууралуу маалымат.

Шайкештик жөнүндө декларация

BM, BMh түрүндөгү соргучтар Бажы Биримдиктин Техникалык регламенттин талаптарына ылайыктуу тастыкталган: TP ТБ 004/2011 «Төмөн вольттук жабдуунун коопсуздугу жөнүндө»; TP ТБ 010/2011 «Жабдуу жана машиналардын коопсуздугу жөнүндө»; TP ТБ 020/2011 «Техникалык каражаттардын электромагниттик шайкештиги».

ARM

BM, BMh

Շահագործման ձեռնարկ

Տվյալ սարքավորման շահագործման ձեռնարկը բաղկացած է մի քանի մասերից.

Մաս 1. սույն «Շահագործման ձեռնարկ»:

Մաս 2. էլեկտրոնային մաս. այն է՝ «Անձնագիր: Մոնտաժման և

շահագործման ձեռնարկ» տեղադրված «Գրունդֆոս». Անցեք փաստաթղթի վերջում նշված հղումով.

Մաս 3. տեղեկություն արտադրման ամսաթվի վերաբերյալ՝ նշված սարքավորման պիտակի վրա:

Տեղեկություններ հավաստագրման մասին՝

BM, BMh տիպի պոմպերը սերտիֆիկացված են համաձայն Մաքսային Միության տեխնիկական կանոնակարգի պահանջների՝ TP TC 004/2011 «Ցածրավոլտ սարքավորումների վերաբերյալ», TP TC 010/2011 «Մեքենաների և սարքավորումների անվտանգության վերաբերյալ» ; TP TC 020/2011

«Տեխնիկական միջոցների էլեկտրամագնիսական համատեղելիության վերաբերյալ»:



<http://net.grundfos.com/gr/i/99133244>

10000113391 0916

ECM: 1193920

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